

MARK 1 EA — Giveback / Butterfly-Effect Investigation

Full Project History, Findings, and Open Threads

Compiled as a handoff record before migrating to a continuous RAG-based research system.

Executive Summary

The project began with a single, real observation: a validated, multi-year-profitable gold (XAUUSD) sweep/momentum EA was losing a large share of its gross profit to trades that reached meaningful floating profit and then reversed into losses ('giveback'). Every attempt to fix this by changing exit management — a fixed R-multiple governor, a 1:1 risk:reward variant, session-scoped rules, and finally a trained machine-learning model — ran into the same underlying obstacle, discovered and proven over the course of the project: any change to when a position closes changes what the EA's single-position architecture searches for next, producing a different trade sequence than the original backtest ('the butterfly / cascade effect'). The project's real, load-bearing achievement is not a finished profitable exit rule — it is a validated, tick-accurate measurement toolchain that finally made this effect provable and quantifiable, and a precise, evidence-based diagnosis of why every exit-side fix hits the same wall.

Phase 1 — Discovering the Giveback Pattern

Starting point: several MT5 Strategy Tester CSV exports of the EA under different configurations (no restrictions, dynamic lot sizing with a max-dollar cap, one-trade-per-day). Direct analysis of these exports (Python, pandas) established the core, real pattern that the rest of the project was built around:

- Across configurations, 74-87% of losing trades had reached at least 0.5R of favorable excursion before reversing to a loss — the strategy was not simply picking bad entries, it was routinely giving back real, already-earned profit.
- Median winning trade peaked far above its eventual close, sometimes 9-10R, indicating the exit logic (fixed take-profit) was not capturing the bulk of favorable moves efficiently.
- Net-to-gross capture ratio (Net Profit / Gross Profit) was consistently only 29-40% across configurations — the strategy was generating far more gross profit than it kept.
- Session analysis showed Asian-session entries — roughly 80% of trade volume — held onto profit far worse than New York entries (New York: ~89% eventual-hold vs Asian: far lower), though this was ultimately set aside since acting on it would itself touch exit timing.

Phase 2 — The Peak-Decay Giveback Governor (Simulated)

Proposed a ratchet-style trailing floor: once a trade's floating profit first reaches an Activation threshold (in R), track its running peak; if profit ever recedes more than a Giveback allowance (in R) from that peak, close the trade. Two parameters, both in R-multiples, both swept across a grid.

Initial validation method: patch the rule onto the already-recorded 5-minute-interval profit snapshots in the existing CSV exports, holding every other trade's entry, timing, and outcome frozen. This is a fast, cheap, but — as later proven — fundamentally limited method.

Result at this stage: extremely optimistic. Governed net profit projected at \$30,000-\$57,000 against a real baseline of roughly \$12,000, across a wide parameter grid, with the governor appearing to leave large winners almost untouched while rescuing the majority of previously-losing trades.

Phase 3 — The Butterfly Effect: Discovery and Proof

The governor logic was implemented directly in the EA's MQL5 source and run as a genuine fresh MetaTrader 5 Strategy Tester backtest — not a patched log. The result did not match the simulation at all:

- Real governed backtest: net profit came in at or below the ungoverned baseline, repeatedly, across many different Activation/Giveback settings — never anywhere close to the \$30-57k projection.
- Trade count changed materially between the baseline and governed runs (e.g. 238 vs 282 trades in one early comparison) even though the entry logic itself was never touched.

Root cause, identified and confirmed directly in the EA's own source code: the EA only searches for a new entry while it is flat (`if(PositionsTotal()>0) return;`), and every entry feature it evaluates (swing high/low, HTF trend state, session, quality score) is computed fresh, from whatever the market looks like at that exact moment. Changing when a position closes — by any amount, for any reason — changes which bar the EA is standing on when it next looks for a signal, which can produce an entirely different trade, or no trade, or a new trade where none existed before. This effect compounds forward for the remainder of the backtest.

This was proven decisively with a direct decomposition: matching a real governed run against the real baseline by entry time and price. A representative result: 203 trades matched exactly, 35 of the baseline's original (good, above-average) trades were crowded out, and 79 new, near-zero-edge trades appeared that never existed in the baseline. The new trades and the lost trades were shown, independently, to explain almost the entire gap between the simulated and real result.

Phase 4 — Attempted Mitigations, All Tested for Real

A series of exit-side ideas were tested, each compiled into the actual EA and run as a genuine backtest rather than trusted as a patched simulation, once the cascade problem was understood:

- **MaxTradesPerDay = 1** (with cooldown and session filter both removed) — the single most effective mitigation found. Verified computationally: zero calendar days ever had more than one trade. This bounds — but does not eliminate — the cascade's exposure, and became the locked-in baseline configuration for all subsequent work. Confirmed stable across the 1-year backtest, landing at \$11,854.77 net — evidence the underlying edge and the giveback pattern are structurally consistent across very different gold regimes (2021 quiet range vs 2024-2026 quadrupled bull market).
- **Fixed 1:1 risk:reward test** — capping every winner at 1R. Rejected: this destroys the trend-continuation upside that is the actual source of the strategy's edge (median winner already runs 9-10R before the fix; verified independently that the live EA's take-profit is fixed at exactly 2.0R to machine precision, so any 'runs to 10R' finding earlier in the project referred to a different, since-superseded configuration).
- **Session-scoped exits (Asian-only 1:1)** — tested as a way to shrink the cascade's blast radius by only touching ~80% of trade volume instead of 100%. Still carries real cascade exposure proportional to how much of the timeline it touches; not pursued further once the measurement tooling below made better options possible.
- **Hour/hour-lock and 'reset the day differently' ideas** — investigated and clarified: the daily trade cap resets on calendar day, but the actual blocking mechanism is 'is a position currently open', not the daily counter. A multi-day-hold trade closing early unlocks entire calendar days that were never searched in the original run, regardless of how the daily counter is defined. This clarified that no daily-reset variant, by itself, can close the gap.

Phase 5 — Building a Tick-Accurate Validation Toolchain

To stop relying on comparing two independently-generated trade logs after the fact, a Python toolchain was built to replay known, already-executed trades against the user's own local real tick archive (HistData-sourced, bid/ask/volume, 1000+ parquet chunk files spanning 2021-2026).

- **Tick chunk indexing** — a cached index of each parquet file's covered date range, so the engine only loads the chunks relevant to a given trade instead of the full multi-gigabyte archive.
- **Timezone/offset auto-detection** — rather than assume the theoretical EST/EET-EEST DST relationship between the tick source and the broker's server time, the engine empirically searched a wide range of hour offsets and picked whichever reproduced real recorded entry prices most closely. Result: a consistent +7 hour offset in both summer and winter samples, average price error under \$0.75 — a real, data-confirmed alignment, not an assumption.
- **Trade replay engine** — replays each known trade (real entry time/price/SL/TP) tick-by-tick from the real archive, rather than re-deriving entry signals from scratch. An earlier full-logic re-implementation attempt was tried first and abandoned after it achieved a 0% match against real results; the replay-only approach, validated against the real baseline, matched within roughly 0.7-1% of real net profit — treated as the trustworthy foundation for everything after.

This toolchain was then used to decompose exactly how much of the gap between the earlier patched-log estimates and real backtests was (a) an artifact of only sampling floating profit every 5 minutes in the original

CSV export, versus (b) the genuine cascade effect. For at least one governor setting, sampling-resolution error was shown to be the dominant term (roughly \$40,000 of a \$40,800 total gap), with the isolated cascade effect measured at only a few hundred dollars for that specific case — an important, non-obvious finding that reframed how much of the earlier disappointment was measurement error rather than the cascade itself.

Phase 6 — The Shadow Signal Logger: Measuring the Parasite Directly

Rather than continue inferring the cascade's size from before/after trade-log comparisons, a 'shadow' detection pass was added to the EA: the full entry-signal logic (swing scan, higher-timeframe filter, structure-break filter, quality score) now runs and is logged on every single bar, completely independent of whether a position is currently open. Only the real order-placement call remains gated on being flat. This is purely observational — it never opens, closes, or modifies a trade — and gives, for the first time, ground-truth visibility into every qualifying signal the EA's own logic ever found, whether or not it was allowed to act on it.

Result, over the full 1-year period: 5,605 qualifying signals found in total.

Outcome	Count	Share
Actually executed (OK)	245	4.4%
Blocked — a position was already open	3,081	55.0%
Blocked — the daily trade quota was already used	2,279	40.7%

- For every 1 real trade taken, roughly 22 more real, quality-filtered signals existed but were blocked purely by timing — a directly measured, not inferred, size for the cascade's raw material.
- Of the position-open-blocked signals, 100% were matched (by timestamp) to the exact real trade that was blocking them. 60.1% occurred within 24 hours of that blocking trade's own entry, and the median missed signal occurred only 755 minutes (12.6 hours) into the blocking trade's life — meaning the sensitivity is not confined to rare, multi-day-hold trades; it is present within hours of the large majority of ordinary trades.
- This reframed the entire problem: the cascade is not a tail-risk edge case, it is close to a structural constant of the current single-position architecture.

Phase 7 — Meta-Labeling: An ML Exhaustion-Probability Layer

Rather than a fixed R-multiple governor, a small logistic regression was trained to predict, for an already-open and already-profitable trade, the probability that it will still close as a real winner — intended as a Tier-1-safe input (sizing/confidence signal) that could eventually inform an exit decision. Trained on tick-replayed checkpoints from the real 1-year trade history, using chronological (not random) train/test splitting to avoid look-ahead.

This went through several real, substantive iterations, each one a genuine bug or design flaw found and fixed, not a cosmetic change:

- **V1 — collinearity bug.** Checkpoints were originally only sampled at the instant a trade first reached a new peak, which made 'drawdown from peak' mathematically zero on every single row by construction. Fixed by adding periodic (every-10-minute) sampling independent of new-peak events, so drawdown could actually vary. This alone raised held-out AUC from 0.661 to 0.861 on the first dataset.
- **Maturity gating.** Live-log validation showed the model was near-coin-flip (AUC 0.511) at the very first checkpoint but usefully predictive (AUC 0.619, real win rate 74-79% at high confidence) by the 8th checkpoint. A maturity gate (require 8 checkpoints of history before trusting or acting on the model) was added and confirmed working exactly as designed in a live re-run.

- **The peak-swamping discovery (V1/V2 failure, found live).** When finally wired to an actual exit action, results were poor — roughly half the damage traced to the real cascade (confirmed by trade-count and matched/lost/new decomposition, consistent with every earlier finding), and the other half traced to a distinct, proven flaw: because the model's *current_R* and *peak_R* features are unbounded and carry large positive coefficients, they mathematically overwhelmed the bounded drawdown penalty once a trade's *R* exceeded roughly 1.0 — proven both algebraically and against the real 1-year log (0% of high-*R* checkpoints ever produced a 'risky' reading under this version, regardless of how much profit had actually been given back).
- **V3 — the structurally sound fix.** Removed every unbounded term from the model, retaining only a bounded, relative *fraction of peak given back* (0-1) alongside velocity, time, hour, and quality score. This was proven mathematically incapable of being swamped by trade size, and confirmed against a fresh live log: peak-size correlation with the model's output dropped to ~0, and genuine low-confidence readings finally appeared at high *R* for the first time.
- **The honest ceiling, found last.** Even with the structural bug fully fixed, real-outcome calibration showed that across the entire statistically meaningful range below the model's 0.5 decision threshold (roughly 3,500 real observations), real eventual win rate never dropped below 50%. The mechanical flaw was real and is fixed; the remaining conclusion is that this specific 5-feature set does not carry enough independent signal to reliably identify trades that are more likely to fail than succeed. This is treated as a genuine, informative negative result, not a coding failure.

Supporting Engineering Notes

- A recurring, load-bearing rule enforced throughout: any exit-timing mechanism must be validated by compiling it into the real EA and running a genuine MT5 backtest — a patched CSV estimate is only ever a cheap, directional pre-check, never a number to act on.
- Multiple real engineering bugs were found and fixed along the way, each confirmed with a targeted test before being handed back: a missing helper function that crashed a diagnostic pass; a dependent helper needing an unusually long (100-hour) price-history window to compute a reliable moving-average baseline; an $O(n^2)$ performance bottleneck in the checkpoint-extraction script (fixed with a constant-time rolling-window structure, verified byte-identical output before and after, roughly 200x faster); a silent date-parsing failure caused by inconsistent timestamp formatting across many small file appends; and a runaway log file caused by a bar-detection guard that only updated on a successful trade rather than on every bar.
- A resume-safe checkpoint mechanism was added to the longest-running script (1-year tick-level feature extraction) after a mid-run interruption, so a rerun continues from where it left off rather than restarting from scratch.

Where This Stands Now

- **Proven and stable:** the giveback pattern is real and consistent across regimes (1-year net profit ~\$11,854.77); the cascade/butterfly effect is real, structurally explained, and measured directly (~5,360 blocked signals against 245 executed trades); the tick-replay validation toolchain is accurate to roughly 1% of real broker output.
- **Tried and concluded (not pursued further):** a fixed-R governor, a 1:1 risk:reward cap, session-scoped exits, and a 5-feature meta-labeling exit model — each tested for real, each understood precisely as to why it did not clear the bar, none abandoned on a guess.
- **Open, unbuilt thread:** a bounded 'fresh-price catch-up window' — when the EA becomes free, checking a short recent window of bars (using the already-built shadow-detection logic) instead of only the single instant-of-freedom bar. This is understood to only recover a fraction of the ~5,360 measured missed signals — those that occur shortly before the EA becomes free — and was paused before sizing the window with real data, since stale prices further back cannot legitimately be executed at all.
- **Also open:** whether a materially richer feature set (tick velocity/acceleration properly normalized, COT-style positioning context, session-specific modeling) could give a meta-labeling approach the signal strength the 5-feature version lacked — not yet attempted.